23 October

Union and Union ALL

Sandton-accounts

account-id	account-holder
A001	Nomvula Dlamini
A002	David Mokoena
A003	Lerato Sithole
A004	Peter Nkosi

Rosebank-accounts

account-id	account-holder
A003	Lerato Sithole
A004	Peter Nkosi
A005	Zanele Khumalo
A006	Thabo Matha

Syntax

```
SELECT account-id,
       account-holder
FROM branch-sandton-accounts
UNION
SELECT account-id,
       account-holder
FROM branch-rosebank-accounts;
```

Output

account-id	account-holder
A001	Nomvula Dlamini
A002	David Mokoena
A003	Lerato Sithole
A004	Peter Nkosi
A005	Zanele Khumalo
A006	Thabo Matha

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Savings-products

product-code	product-name
SV01	Basic savings
SV02	Premium savings
SV03	Youth savings
SV04	Business savings

Current-products

Product-code	product-name
CR01	standard current
CR02	Gold current
SV03	Youth savings
CR03	Business current

Syntax

```
Select product_code,
       product-name
FROM  savings-products
UNION
SELECT product-code,
       product-name
FROM  current-products;
```

Output

product-code	product-name
SV01	Basic savings
SV02	Premium savings
SV03	Youth savings
SV04	Business savings
CR01	Standard Current
CR02	Gold current
CR03	Business current

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retail_banking_staff

staff_id	staff_name
S001	Mpho Radebe
S002	Brian Tshabalala
S003	Aisha Patel
S004	Kabelo Moabelo

corporate_banking_staff

staff_id	staff_name
S003	Aisha Patel
S005	Nandi Dube
S006	Sipho Khumalo
S004	Kabelo Moabelo

Syntax

```
SELECT staff_id,
       staff_name
FROM retail_banking_staff
UNION
SELECT staff_id,
       staff_name
FROM corporate_banking_staff;
```

Output

staff_id	staff_name
S001	Mpho Radebe
S002	Brian Tshabalala
S003	Aisha Patel
S004	Kabelo Moabelo
S005	Nandi Dube
S006	Sipho Khumalo

4. mobile-branch-cities

city-code	city-name
C01	Johannesburg
C02	Pretoria
C03	Capetown
C04	Durban

digital-branch-cities

city-code	city-name
C03	Capetown
C05	Polokwane
C06	Port Elizabeth
C01	Johannesburg

Syntax

```
SELECT  city-code,
        city-name
FROM  mobile-branch-cities
UNION
SELECT  city-code,
        city-name
FROM  digital-branch-cities;
```

Output:

city-code	city-name
C01	Johannesburg
C02	Pretoria
C03	Capetown
C04	Durban
C05	Polokwane
C06	Port Elizabeth

5 push-notification-targets

customer-id	customer-name
C1001	Nomsa Zwane
4002	Andile Buthelezi
C1003	Fatima Mahomed
C1004	Ryno van Zyl

inapp-banner-targets

customer-id	customer-name
C1003	Fatima Mahomed
C1005	Thandeka Cele
C1006	Samuel Nkosi
C1002	Andile Buthelezi

Syntax

```
SELECT  customer-id,
        customer-name
FROM push-notification-targets
UNION
SELECT  customer-id,
        customer-name
FROM  inapp-banner-targets;
```

Output:

customer-id	Customer-name
C1001	Nomsa Zwane
C1002	Andile Buthelezi
C1003	Fatima Mahomed
C1004	Ryno Van Zyl
C1005	Thandeka Cele
C1006	Samuel Nkosi

6 atm01-transactions

txn-id	account-id	amount
T1001	A001	500.00
T1002	A002	1200.00
T1003	A001	300.00
T1004	A003	750.00

atm02-transactions

txn-id	account-id	amount
T1003	A001	300.00
T1005	A004	900.00
T1006	A002	450.00
T1007	A005	150.00

Syntax

```
SELECT  txn_id,
        account-id,
        amount
FROM    atm01-transactions
UNION ALL
SELECT  txn-id,
        account-id,
        amount
FROM    atm02-transactions;
```

Output:

txn-id	account-id	amount
T1001	A001	500.00
T1002	A002	1200.00
T1003	A001	300.00
T1004	A003	750.00
T1003	A001	300.00
T1005	A004	900.00
T1006	A002	450.00
T1007	A005	150.00

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gauteng-loan-applications

app-id	customer-id	loan-type
LA001	C1001	Home loan
LA002	C1002	Vehicle
LA003	C1003	Personal
LA004	C1004	Home loan

Western-cape-applications

app-id	customer-id	Loan-types
LA003	C1003	Personal
LA005	C1005	Vehicle loan
LA006	C1006	Business
LA007	C1007	Home loan

Syntax

```
SELECT  app-id,
        customer-id,
        loan-type
FROM gauteng-loan-applications
UNION ALL
SELECT  app-id,
        customer-id,
        loan-type
FROM  western-cape-applications;
```

Output

app-id	customer-id	loan-type
LA001	C1001	Home loan
LA002	C1002	vehicle
LA003	C1003	Personal
LA004	C1004	Home loan
LA003	C1003	Personal
LA005	C1005	Vehicle loan
LA006	C1006	Business
LA007	C1007	Home loan

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email-complaints

complaint-id	customer-id
ECOO1	C2001
ECOO2	C2002
ECOO3	C2003
ECOO4	C2004

app-complaints

complaint-id	customer-id
ACOO1	C2005
ACOO2	C2001
ACOO3	C2006
ACOO4	C2007

Syntax

```
SELECT complaint-id,
       customer-id
FROM email-complaints
UNION ALL
SELECT complaint-id,
       customer-id
FROM app-complaints;
```

Output

complaint-id	customer-id
ECOO1	C2001
ECOO2	C2002
ECOO3	C2003
ECOO4	C2004
ACOO1	C2005
ACOO2	C2001
ACOO3	C2006
ACOO4	C2007

9 april-payment

payment-id	account-id	amount
PAY001	A001	12500.00
PAY002	A002	4800.00
PAY003	A003	9200.00
PAY004	A004	3300.00

may-payment

payment-id	account-id	amount
Pay 005	A001	12500.00
Pay 006	A005	7600.00
Pay 001	A002	5100.00
Pay 008	A006	2800.00

Syntax

```
SELECT payment-id,
       account-id,
       amount
FROM april-payment
UNION ALL
SELECT payment-id,
       account-id,
       amount
FROM may-payment;
```

Output

payment-id	account-id	amount
PAY001	A001	12500.00
PAY002	A002	4800.00
PAY003	A003	9200.00
PAY004	A004	3300.00
PAY005	A001	12500.00
PAY006	A005	7600.00
PAY007	A002	5100.00
PAY008	A006	2800.00

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debit_entries

entry_id	account_id	entry
DR001	A001	Debit
DR002	A002	Debit
DR003	A003	Debit
DR004	A004	Debit

credit_entries

entry_id	account_id	entry
CR001	A001	Credit
CR002	A005	Credit
CR003	A002	Credit
CR004	A006	Credit

Syntax

```
SELECT entry_id,
       account_id,
       entry
FROM debit_entries
UNION ALL
SELECT entry_id,
       account_id,
       entry
FROM credit_entries;
```

Output

entry_id	account_id	entry
DR001	A001	Debit
DR002	A002	Debit
DR003	A003	Debit
DR004	A004	Debit
CR001	A001	credit
CR002	A005	credit
CR003	A002	credit
CR004	A006	credit

11 Union combines the results from January and February high-risk watch lists and removes duplicate customers it gives fraud analysts a unique list of customers who appeared in either month

12. Union ALL combines records from different queries and keeps all records, including duplicates, since the audit team wants to count every individual transaction they must not remove duplicate records

13 Union requires both queries to return the same number of columns, in the same order, with compatible data types